

WeedDiffusion: A Dual-Branch Synthetic Augmentation Framework for Weed Mapping

Pasquale De Marinis, Antonio Iammarino, Gennaro Vessio, Giovanna Castellano



Motivation

- ▶ Weeds cut crop yields by **25 to 80%**.
- ▶ Pixel-level labels for segmentation are **costly and slow** to produce.
- ▶ Standard augmentation adds only **surface variation**, not new content.
- ▶ Small, imbalanced datasets **overfit** on the rare weed class.

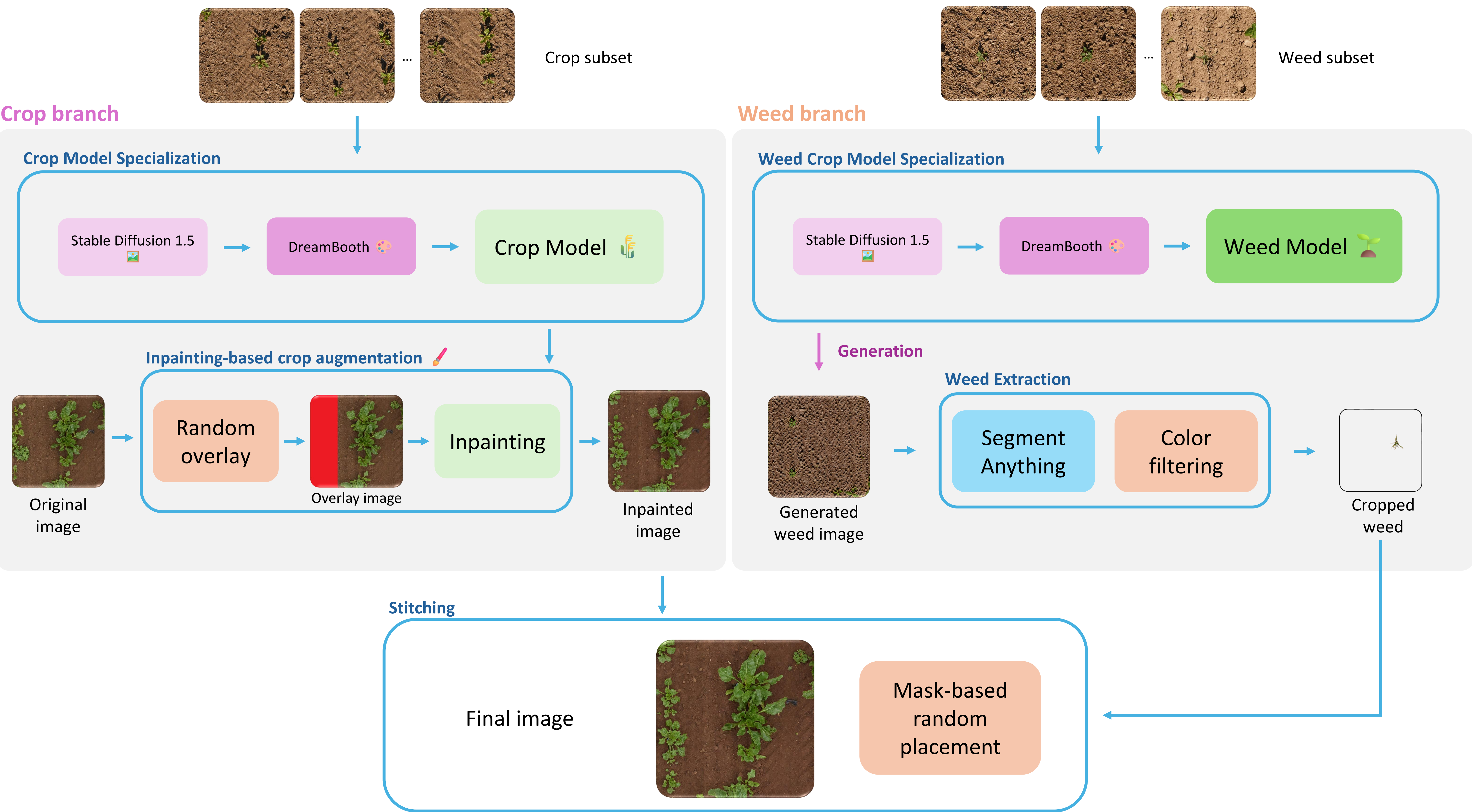
What we propose

WeedDiffusion, which fine-tunes Stable Diffusion v1.5 with DreamBooth [1] to generate labeled crop and weed data through two branches:

- ▶ **Crop branch**: inpaints new crops into real fields.
- ▶ **Weed branch**: generates weeds, cuts them out with SAM, and pastes them into backgrounds.

From 37 real images, we produced 120 synthetic samples that improve mIoU across ERFNet, DeepLabV3+, and UNet, most on the weed class.

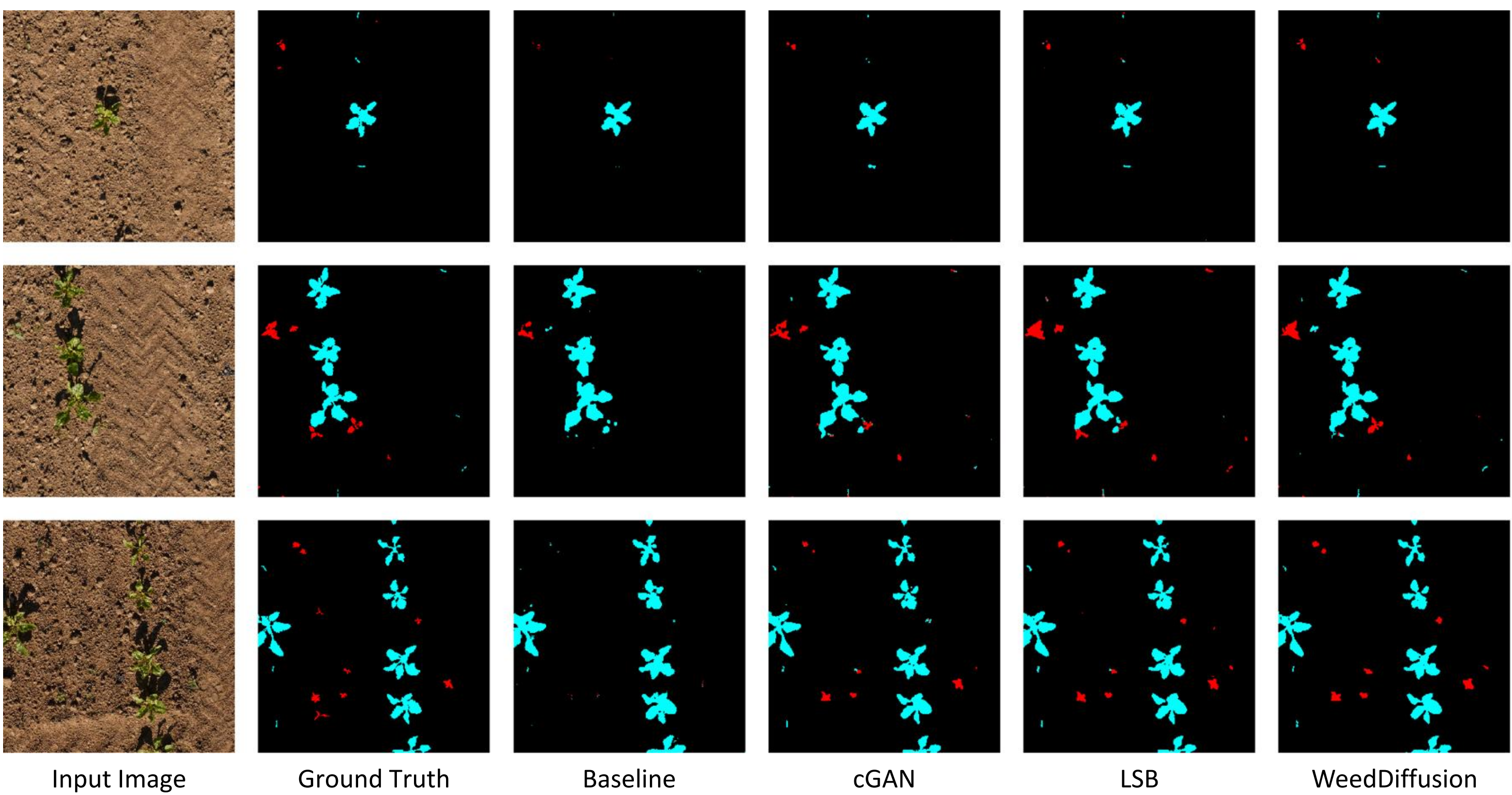
Method



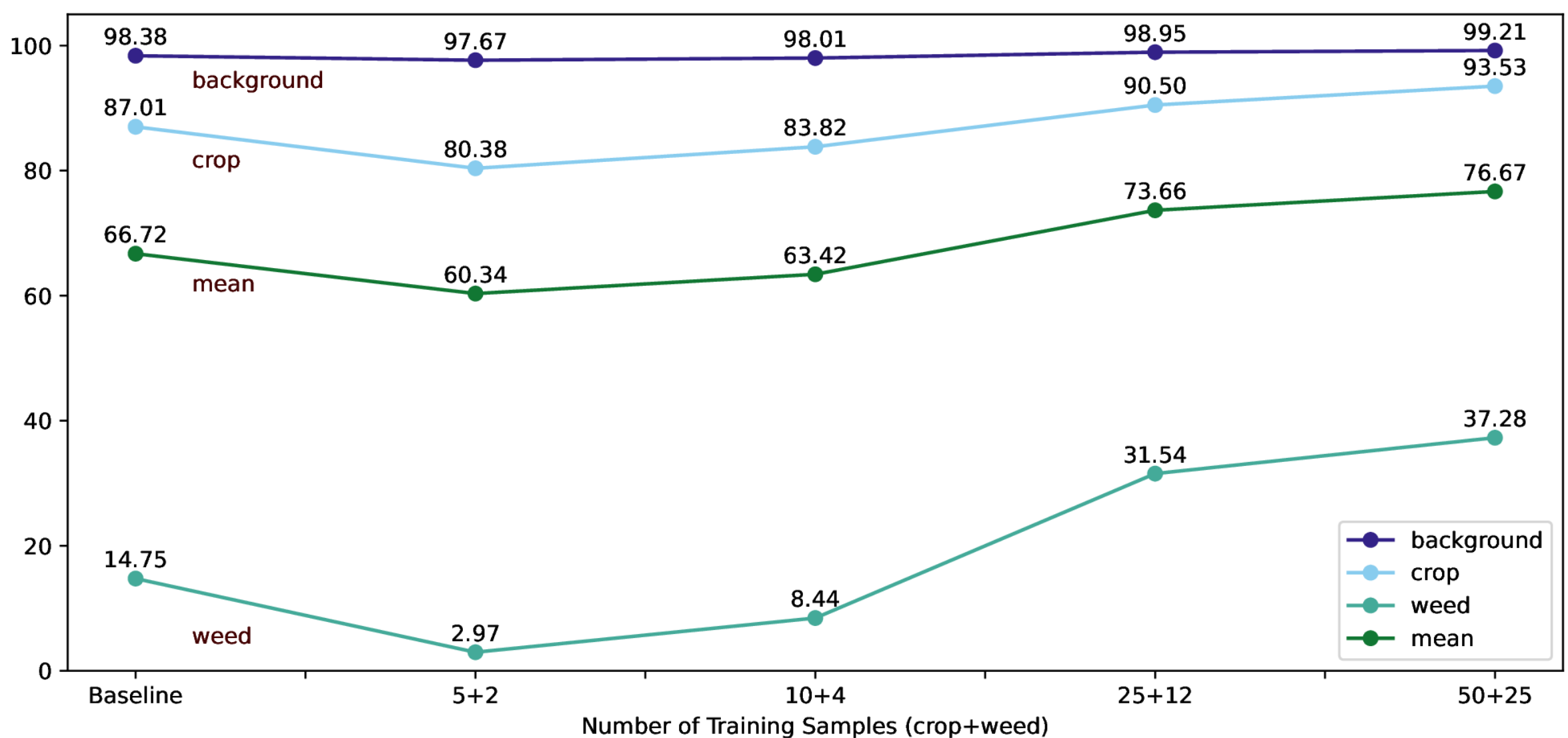
Segmentation quantitative results

Architecture	Online aug.	Offline aug.	BG	Crop	Weed	Mean
ERFNet	geo+color	none	98.38	87.01	14.75	66.72
	geo+color	cGAN	98.51	87.04	21.34	68.96
	geo+color	LSB	98.74	87.89	22.57	69.73
	geo+color	WeedDiffusion	98.95	90.89	40.02	76.62
DeepLabV3+	geo+color	none	98.52	86.18	10.14	64.95
	geo+color	cGAN	98.99	87.35	17.68	68.00
	geo+color	LSB	99.02	87.85	19.43	68.77
	geo+color	WeedDiffusion	99.04	91.10	35.16	75.10
UNet	geo+color	none	99.05	91.02	29.98	73.35
	geo+color	cGAN	99.13	91.45	30.87	73.82
	geo+color	LSB	99.17	91.83	31.54	74.18
	geo+color	WeedDiffusion	99.26	92.90	45.55	79.23

Segmentation qualitative results



Data efficiency



Conclusion

- ▶ WeedDiffusion generates labeled crop and weed data from **only 37 real images**.
- ▶ It **raises mIoU** by 5.9 to 10.1 points across ERFNet, DeepLabV3+, and UNet.
- ▶ The **weed class gains most**, with IoU more than doubling in every model.
- ▶ The approach is **model-agnostic** and works well in low-data settings.

[1] Ruiz et al. DreamBooth: Fine-tuning text-to-image diffusion models for subject-driven generation. CVPR, 2023.



Project Page



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